The Internet and World Wide Web (WWW)

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Overview

- The Internet
  - Meaning of the internet
  - Why connecting computers
  - What you can do with internet
  - Pros and cons of the internet
  - Types of network
  - Things required to connect to the internet
- The web
  - WWW
  - Hyperlink
  - Web page
  - Website
  - URL
  - Web browser

Meaning of the internet

- It is the interconnection of computer networks all over the world so that users can share and exchange information.
- A computer network is a connection of two or more computers for the purpose of communicating and sharing information and other resources.

➢Computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

  ➢through cable, telephone lines or through wireless communication.

Why connecting computers?

- The main aim of connecting these computers is to enable sharing of resources such as a hard disk, printers, documents, computer programs and modems

What can you do with the internet?

- Find information by using search engines
- File Transfer
- International debates or discussion groups
- Send messages by e-mail
- School (eLearning)
- Electronic Trading
  - Companies and their customers making use of networks as a means of advertising, buying and selling goods and services
- Teleworking or Telecommuting
  - Making use of networking technology to allow working ad different location while keeping in touch with co-workers
- Tele and video conferencing
  - Groups of people conducting meetings via a telephone or video network.

What can you do with the Internet?

- Electronic Mail
  - Permits you to send and receive mail and provides access to discussion groups often called listserv.
- File Transfer Protocol
  - Allows your computer to rapidly retrieve complex files intact from a remote computer and view or save them on your computer
- Publish information on your own web site
- Join a club through a newsgroup
- Find someone in an online phone book
- Have a conversation in a 'chat' room
- Make airline reservations, invest in the stock market
- Get directions and draw maps
- Go shopping and more!

Benefits of using the Internet

- Globalization
  - The Internet has given us the opportunity to connect to any place in the world.
- Enhance business
  - E-commerce websites, E-mails
- Connecting people
  - Social networks i.e. Facebook, WhatsApp, Instagram, twitter
- Faster and easier source of information
  - Compared to the physical library
- Enhanced education approach
  - E-learning (Online classes, tutorials, online courses, books accessed online)

Limitations of using the Internet

- Problematic addiction
  - People spend much time on the internet other than with friends
- Loss of culture
  - People are losing their social characteristics, they are adapting them to indirect communication on the Internet
- Lack of privacy
- Dangerous possibility of stealing money, information and identity,
- Illegal distribution of music and videos
  - Eg pornography

Types of network

- There are many types or groups of networks.
  - Personal Area Network (PAN)
  - Local Area Network (LAN)
  - Wide Area Network (WAN)

Local Area Network

- This is an interconnection of computers and other devices within a single geographical site.
  - e.g within an office, a building or a complex of buildings within a campus.
  - A good example a Mzuzu University ICTLAB
- LAN is a private network.
- Do not require external network services e.g. telephone network or leased data lines.
- LAN are widespread as a result of:
  - Increasing need to transfer data between computers within the organization.

Purpose of LAN

- LANs allow:
  - Data transfer between computers
  - Access to data stored somewhere
  - Messages to be passed between computers
  - Shared use of peripherals e.g. printers
  - Shared use of computing resources

Wide Area Network (WAN)

- Covers a large geographical area to provide national and international communication.
  - This is a network that connects computers in different locations either regional, national or International.
- Connections require the use of Telecommunication equipment such Modems, Telephone lines satellites or wireless communication.
- WANs can be accessed through dial-up connections, using a modem or leased-line direct connection.

What is needed to connect to the Internet?

- Hardware
  - Computer, Modem, Dongle, Telephone line, Wireless infrastructure, cables, hub, switch, routers etc.
- Internet Service providers (ISP)
  - **ISP** refers to a company that provides Internet services, for personal and business access to the Internet e.g. MTL, Airtel, TNM, Globe Internet etc..
- Software
  - Web Client Software (Browsers)
- Protocols
  - IP, TCP/IP, http, ftp and many more

The World Wide Web (WWW, or The web)

- **World Wide Web** is a collection of electronic documents that are linked together like a spider web.
- The Web is part or service of the Internet that uses text, images, sound, and video.
- These documents are stored on computers called **servers** located around the world.
- Information is presented on a web page one at a time and each page contains links called hyperlinks
  - These hyperlinks could be words, pictures or graphics
- This linking is called hypertext.

WWW

- It is called a web because if you join up all the links on the pages on all the computers on all networks on the Internet, it would just look like one big mesh web
- A page of text which contains hyperlinks is called **hypertext**
  - ***Hypertext*** is text which contains links to other texts.
- **Surfing** or **browsing** is navigation through pages of information.

Hyperlinks

- A **hyperlink** is a component of a web page that allows quick navigation through a website or to a new website.
- Hyperlinks are contained in web pages
- The mouse will turn into a hand shape when moved over a link.
- With hyperlinks one can move through a vast amount of information by jumping from one web page to another.
- Hyperlinks connects you to other pages of related information
- For example you click on a link of Administration of Mzuzu University Home Page, you will land on administration page

Hyperlinks

- Clicking on a hyperlink may take you to a page of information which resides on the same computer or to a page which is stored on a computer on the other side of the world.

Web page

- A **web page** is an electronic document or resource of information that can be accessed through a web browser.
- A web page is written in a computer language called HTML.
- A web page is an individual page that can be found on a web site.
- Web pages are stored on web servers located around the globe
- A web page is an electronic document on the world wide web.
- A web page contains many things:
  - Information: Graphics, sound, movies, text, animations.
  - Other features: links, forms
- A web page contains links to other web pages as well.

A Website

- A **website** is a collection of mostly similar web pages
  - The web pages are based on a common theme eg a person, business, organization, or a subject like sports.
- A web site is similar to a magazine with articles. It is a collection of web pages.
- A web page is an individual page that can be found on a web site.
- Most websites have a home page as their starting point, which frequently functions as a table of contents for the site.

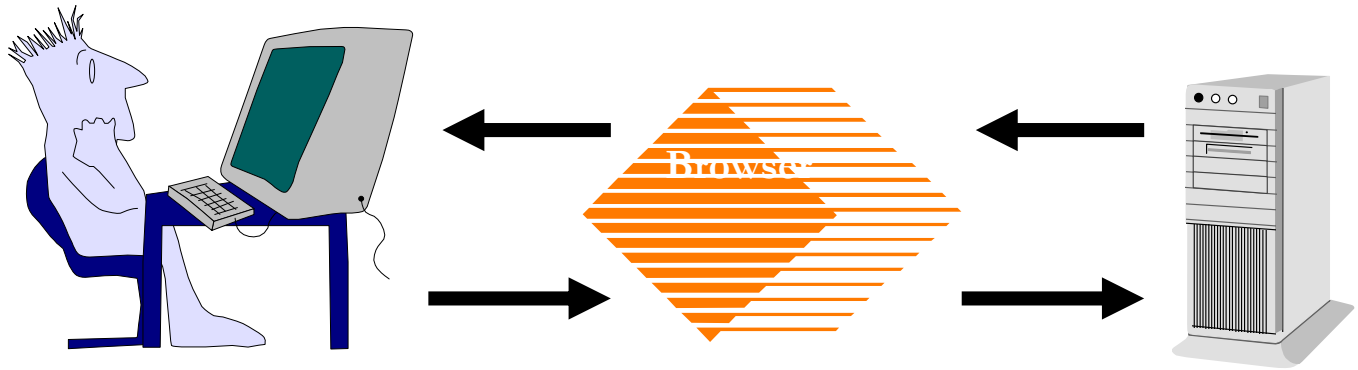
What is a Web Browser

- A Web browser is a software that is used to find, retrieve, view, and send information over the Internet.
- User enters an address (URL) into the browser's address bar or search box.
- The browser communicates with the server to get the information requested by the user.
- This software allow users to browse – or search - the World Wide Web.
- Examples of Web Browsers include:
  - Mozilla Firefox, Google Chrome, Mosaic, Netscape Navigator, Internet Explorer, opera, safari etc.

How the Web Works

5. User receives file displayed by the browser

4. Server sends requested files to browser to be interpreted



1.User sends request

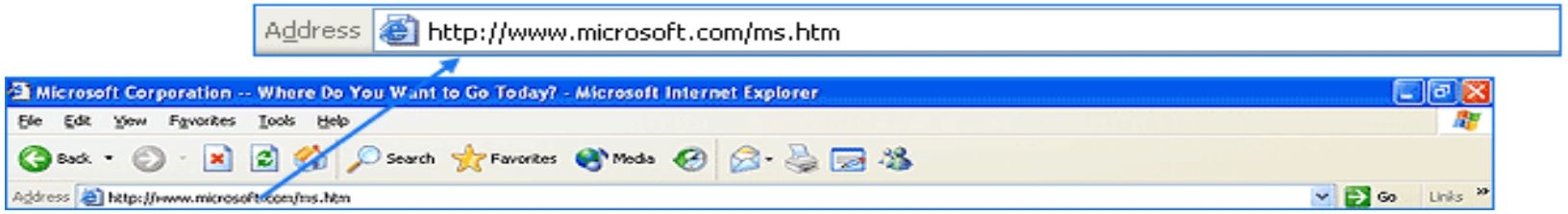
2. Browser interprets user's selection and makes request from appropriate server

3. Server accepts and processes request from browser

Web Server

- **Web Server** is a computer that provides world wide web services on the Internet.
- Or a **web server** is a computer that accepts client requests and responds to them
- It includes the hardware, operating system, web server software, TCP/IP protocols and the web site content (Web pages).

URL



- A URL (Uniform Resource Locator) is the address of an Internet page on a Web site. E.g. http://www.mzuni.ac.mw
- It indicates where a web page is located on the internet.
- A location identifier associated with a particular webpage or file on the Internet, e.g. www.yahoo.com
- URL is typed in the address bar of the browser.

- URL is the address that defines the route to a file on the Web.
- URL are typed into the browser to access web pages
- URL contains the protocol prefix, domain name, subdirectory name and file name
- To get access to a home page on web site, only the protocol and domain name are required for example:
- http://www.mzuni.ac.mw
  - A **domain name** is a unique name that identifies an internet resource such as a website
  - A part of an email or website address on the Internet that shows the name of the organization that the address belongs to

- The URL specifies the Internet address of a file stored on a host computer connected to the Internet.
- Every file on the Internet has a unique URL.
- Web software programs use the URL to retrieve the file from the host computer and the directory in which it resides
- This file is then displayed on the monitor connected to the user's local machine

Format of URL

- Each part of the URL directs the browser to the correct web page.



Different Domains on the WWW

.com - commercial

.net - network

.edu - educational

.org - organization

.gov - government

Protocols

- A set of rules that enable two devices to connect and transmit data to one another.
- A computer language allowing computers that are connected to each other to communicate

✓Protocols determine how data are transmitted between computing devices and over networks. Below are some of them:

TCP /IP

- i)Transmission Control Protocol.
  - This is the standard protocol computers utilize to communicate with one another over a network.
- ii) Internet Protocol
  - ✓This a protocol that enables the computers to identify each other over the Internet

HTTP

✓Hyper Text Transfer Protocol

- A protocol for informing web browsers how to display information found on a particular webpage.
- http is the main protocol used by the World Wide Web to define how messages are formatted and transmitted.
- All Web Pages on the World Wide Web use HTTP, making them accessible to all other Web computers

FTP

✓File Transfer Protocol.

- This is a protocol computers use to send files to one another. You may either upload or download files using FTP
  - Various types of files (executable programs, graphics, audio, text)

Upload

- To forward a file to a server for later access over the Internet.

Download

- To copy a file from an Internet server to a recordable media, such as a hard drive, USB disk or a floppy disk.

SMTP

✓Simple Mail Transfer Protocol

- is a protocol used in sending e-mail

POP3

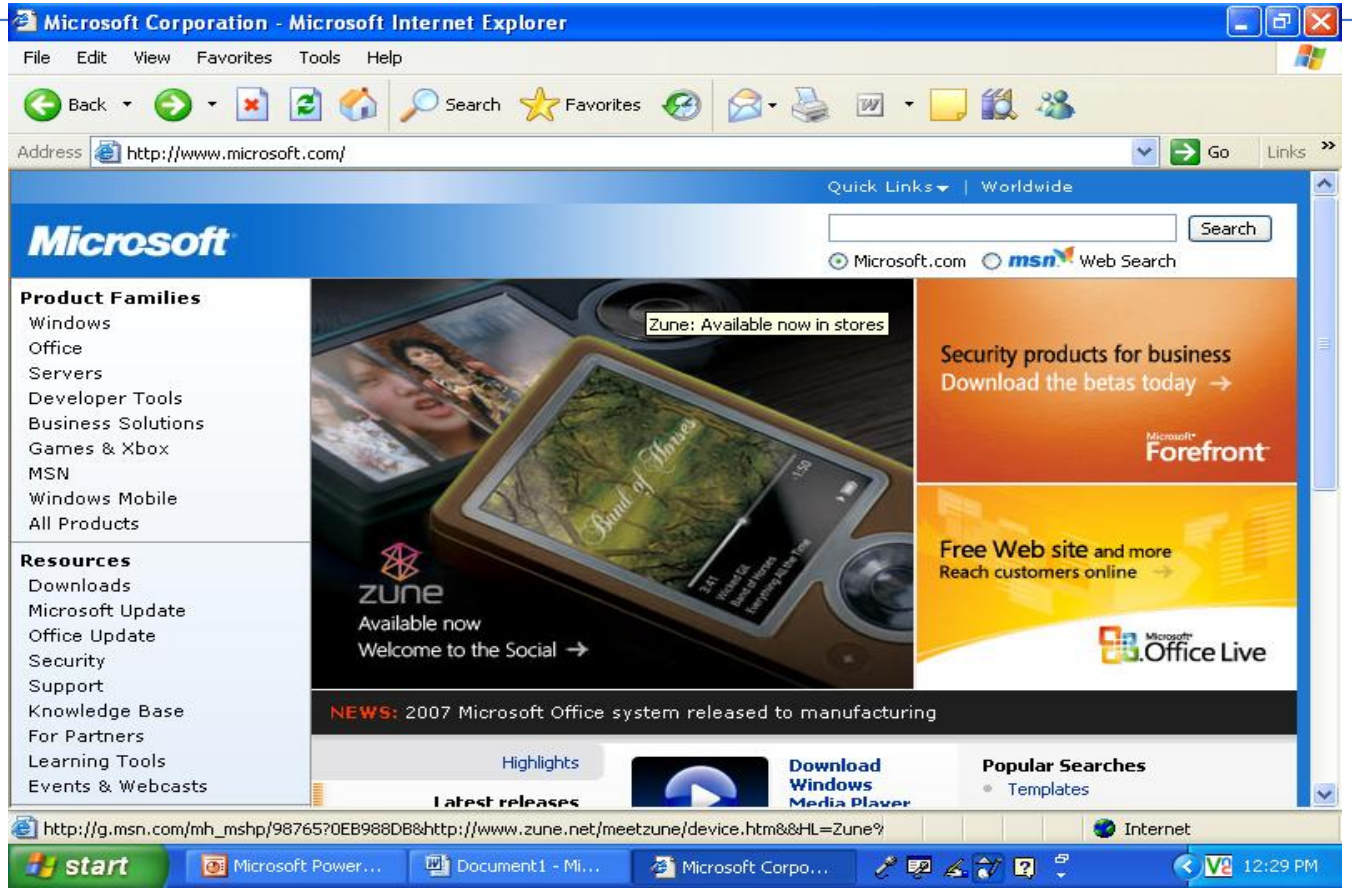
✓Post Office Protocol 3

- a standard protocol for receiving e-mail. Mail is downloaded to your computer (can keep a copy on the server)

Browsers: Microsoft Internet Explorer

- Microsoft Internet Explorer is the most popular web browser.
- It comes with windows operating system.
- When you start Internet Explorer, the Internet Explorer's main browser window opens automatically
- To display a web page you want, you need to know the website address of the web page you want to view

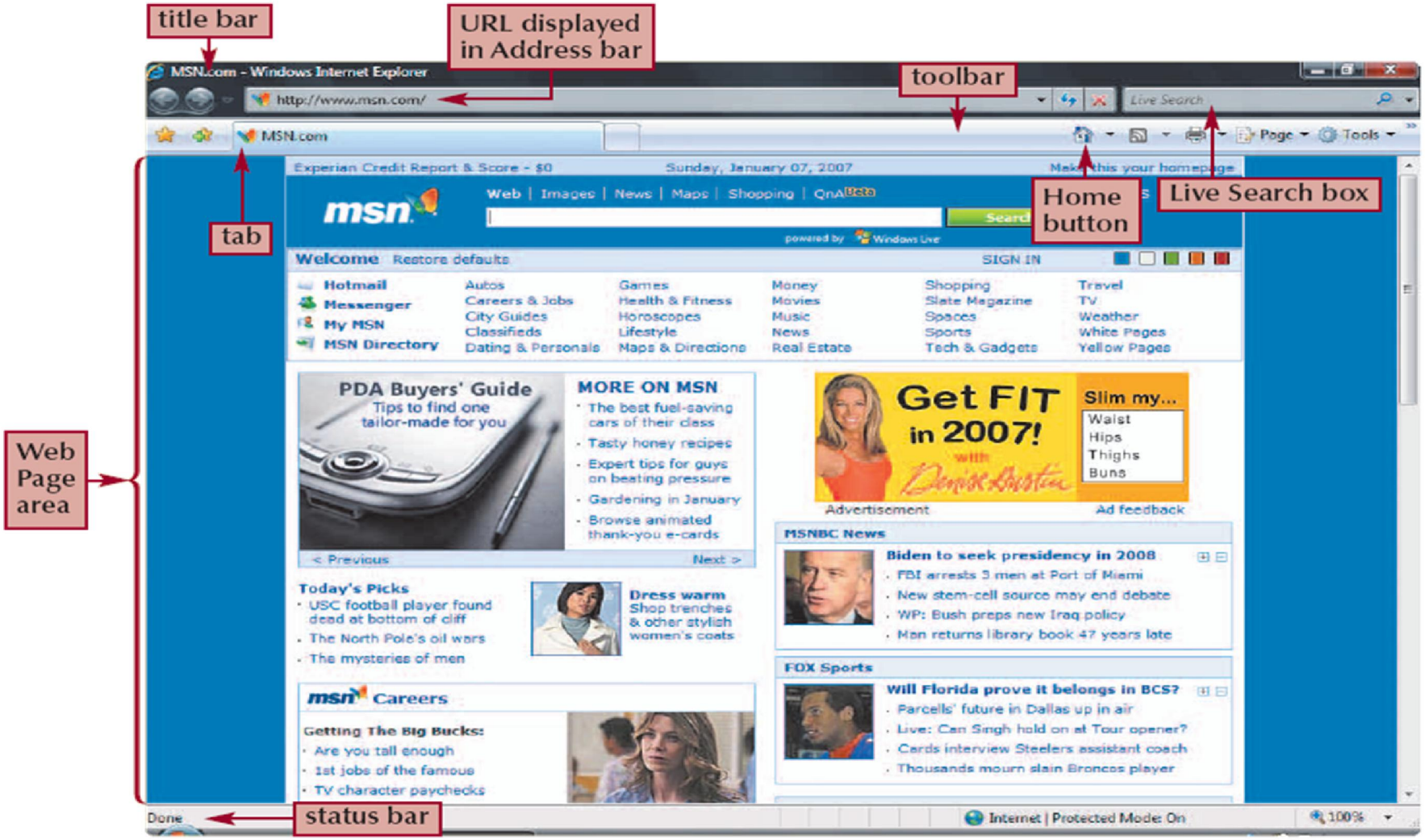
Microsoft Page



Getting Started with Microsoft Internet Explorer

- Click **Internet Explorer** icon on the taskbar

Internet Explorer window **Figure 4-3**



Parts of a Web Browser

- **Title Bar:** Lists the title of the web page that you are viewing. There are three buttons to the right of the title bar which reduce, maximize, or close the screen.
- **Content Area:** Displays the current web page that is open on the browser.
- **Address Field:** Shows the URL (Web address) of the web page that is currently displayed.
- **Status Indicator:** Shows whether or not the Web page has completely loaded and can be viewed.

- **Progress Bar:** Indicates how much of the web page has loaded for viewing.
- **Status Message Field:** Tells you the status of the web page. Indicates when the web page is done and open for viewing.
- **Scroll Bar:** This is used to move up or down on the web page.

How to Access Web Pages

Here are three ways to access a web page:

1. Type the Uniform Resource Locator (URL) in the address box.
2. Click on hypertext links on a web page.
3. Use a search tool such as a keyword search engine or web directory.

Using the URL in the Address Box

- Type the Uniform Resource Locator (web address) in the browser address box.
- Click GO (or press the ENTER button on the keyboard).
- The web address must be correct (no misspellings, extra spaces, etc.) to go to the correct web page.

Opening a Page on the Web

- Use the Back and Forward buttons to navigate through the pages you have visited
- To open a Web page you visited in the last 20 days, use the History list
- To open a new Web page without closing the one you're currently viewing, use **tabbed browsing**
  - If you have two or more tabs open, use the Quick Tabs button to display a thumbnail of each Web page

Displaying History List

- The Internet Explorer has a history list to keep track of the web pages you have recently visited
  - Or redisplay a web page that you visited previously
- Click on history button to display a list of the web pages you have recently viewed
- The History list appears, displaying a list of the web pages you have recently viewed
- Click on the week or day you viewed the web page you want to view again

- The Web site folder with the files you viewed during the week during the week or day appear
- Click on the web site of interest
- The web pages you viewed on the web site appear
- Click on the Web page you want to view
- The web page appears
- When you finish using the History list, click on the History button again to hide it.

About Favorites

- You can make use of the favorites feature when you visit a web site very often.
- You can display your favorites web page without writing the web site address each time you visit a site.
- Steps
  - Display the web page you want to add to your list of favorite web pages
  - Click on Favorites on the Menu bar
  - Click on Add to Favorites Option from the Favorites menu
  - Add Favorite dialog box appears
  - The Name of the web page appears in the Name: Box area

Adding to my Favorites

- Click on the OK button to add the web page to your list of favorites
- The web site address appears in the Favorites menu.
- Whenever you want to visit that site, simply click on the Favorites menu and then click on the site in that menu